

CISCO PACKET TRACER

BANK FACILITY NETWORK DESIGN

PHYSICAL & LOGICAL TOPOLOGY | VLAN SEGMENTATION | LAYER 2 & LAYER 3
REDUNDANCY | ROUTING | SECURITY & SERVICES



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RELIABLE

AUTHORS

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DESIGNING TODAY
FOR A STRONGER
CONNECTED TOMORROW

*Built for
Better Banking*



Document Control

Field	Value
Document	Bank Main Branch — Network Design Master Document
Cover title	Bank Facility Network Design (Cisco Packet Tracer)
Project	Bank Main Branch Network Design (Packet Tracer)
Authors	Innocent Nkosiye Mbatsha (L3 & L7) · Nondumiso Mbuyazi (L2 & L3)
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1 Project Information

1.1 Project brief

The Bank Main Branch network is a **three-tier campus** built in Cisco Packet Tracer. Six access switches (ASW1–ASW6) in three IDFs connect to a redundant distribution pair (L3SW1/L3SW2) in the MDF. The pair provides SVI gateways with HSRP and is joined by a routed L3 EtherChannel (Po1). Router0 is the campus edge and the only DHCP server. A hop-by-hop static path (Router0 → Router1 → Bank-Edge → ISP → HQ Edge) links the branch to the **Head-Quarters LAN 172.16.10.0/24** and its server, Server0 (172.16.10.10).

Field	Value
Document	Bank Main Branch — Network Design Master Document (combines the L2 command book and the L3 & L7 configuration report)
Authors	Innocent Nkosiye Mbatia (L3 & L7) · Nondumiso Mbuyazi (L2 & L3)
Topology of record	Packet Tracer canvas 2026-09-30 (Section 4). Configs are target state; items marked VERIFY must be confirmed in the running .pkt

1.2 Scope

In scope	Not in this revision
L1 devices, models, cabling and port map · floor plan, MDF and IDF placement	Dynamic routing (OSPF/EIGRP) — the design uses static routes only
L2: VLANs, access ports, 802.1Q trunks (native 99), PortFast, BPDU Guard, L3 EtherChannel (LACP)	ACLs, NAT and firewalling
L3: SVIs, HSRP, static routing inside the branch and to HQ	Wireless SSID/security settings (APs are placed on VLANs only)
Services: DHCP (Router0) with relays, DNS (10.10.90.10) · test plan · troubleshooting log · configs	Internet access — 8.8.8.8 is a secondary DNS entry with no path in the lab

1.3 Goals

- **Segmentation** — one VLAN and /24 per department (10 user VLANs), trunks carry native VLAN 99 only for untagged traffic.
- **Gateway redundancy** — every user VLAN uses HSRP VIP 10.10.<vlan>.1; L3SW1 Active, L3SW2 Standby, both preempt.
- **Automatic addressing** — one DHCP server (Router0) and `ip helper-address 10.10.255.1` on both Dist switches.
- **Name resolution** — DNS server 10.10.90.10 handed out by every pool.
- **HQ reachability** — static routes in both directions on every hop to 172.16.10.0/24.
- **Edge protection** — PortFast + BPDU Guard on end-device ports Fa0/1–20 only.
- **Evidence** — every claim is closed with a test result or screenshot (Sections 9–10), not with this document.

1.4 Divided responsibilities

Owner	Responsibilities	Sections
Nondumiso Mbuyazi L2 & L3	VLANs · SVIs · trunks · PortFast · BPDU Guard · L3 EtherChannel (LACP) · VLAN colour scheme · end-device organisation · floor-plan integration	2, 5, 6, 7, A.4–A.5
Innocent Nkosiye Mbatia L3 & L7	DHCP · HSRP · static routing (internal and to HQ) · DNS	3, 5, 8, 9, 10, A.1–A.4

1.5 Design lock

Item	Locked value
VLANs	User VLANs 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 → 10.10.<vlan>.0/24 · VLAN 99 = native on trunks only (no SVI, HSRP or DHCP)
Access	ASW1–ASW6 (2960-24TT) · IDF A = ASW1–2, IDF B = ASW3–4, IDF C = ASW5–6 · end devices on Fa0/1–20 · Gi0/1 → L3SW1, Gi0/2 → L3SW2
Distribution	L3SW1/L3SW2 (3650) · Gi1/0/3–8 → ASW1–6 · Po1 = Gi1/0/23–24, routed, LACP, 10.255.255.0/30 (.1/.2) · SVIs .2 / .3
HSRP	Group = VLAN ID · VIP .1 · L3SW1 priority 110 + preempt (Active) · L3SW2 priority 100 + preempt (Standby)
DHCP	Router0 only · one pool per user VLAN · default-router = VIP · excluded .1–.10 (VLAN 90 .1–.49) · helpers 10.10.255.1 on both Dist
DNS	dns-server 10.10.90.10 8.8.8.8 on every pool · DNS server 10.10.90.10 (VLAN 90)
Routing	Static only · L3SW1 default → 10.10.255.1 · L3SW2 default → 10.255.255.1 · 172.16.10.0/24 outbound and 10.10.0.0/16 return on every hop
HQ	HQ LAN 172.16.10.0/24 · Server0 172.16.10.10 · HQ Edge LAN gateway 172.16.10.1 (VERIFY)

2 Floor Plan and Physical Placement

2.1 Ground-floor plan

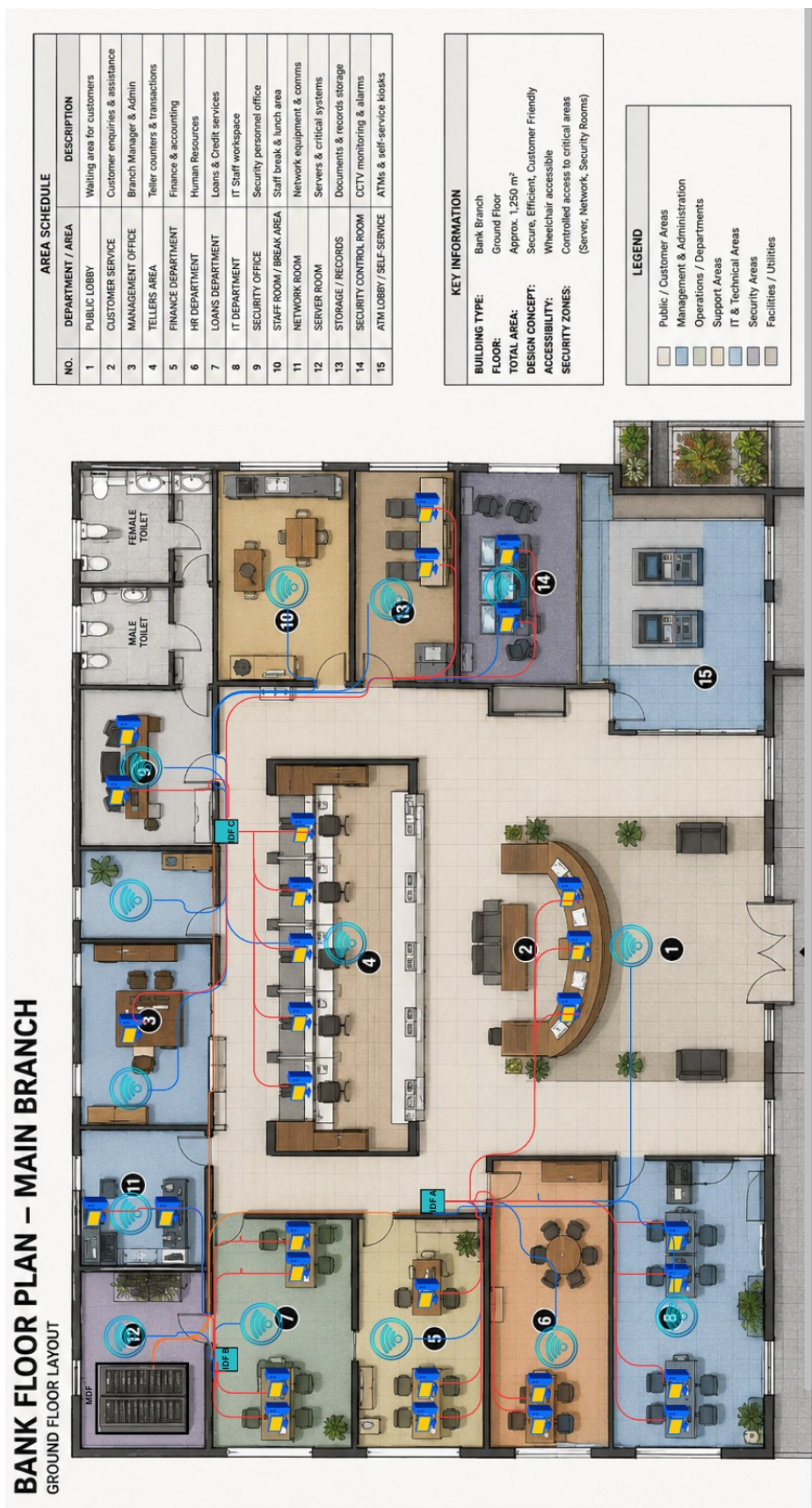


Figure 1 — Bank Main Branch ground floor ($\approx 1250 \text{ m}^2$): 15 rooms with one AP each, MDF in Room 12 (Server Room), IDF A, IDF B and IDF C racks. Rotated 90°: turn the page clockwise to read.

2.2 Rooms, VLANs and IDF s

Room	Name	VLAN	IDF	Access switch ports / notes
1	Public Lobby	— (none assigned)	—	Open item V9: suggest Guest 100
2	Customer Service	60 CUSTOMER_SERVICE	IDF A	ASW1 Fa0/15–17 · ASW2 Fa0/13–15
3	Management Office	10 MANAGEMENT	IDF C	ASW5 Fa0/6–9 · ASW6 Fa0/1–3
4	Tellers	50 TELLERS	IDF C	ASW5 Fa0/1–5 · ASW6 Fa0/4–5
5	Finance	20 FINANCE	IDF A	ASW1 Fa0/1–3 · ASW2 Fa0/1–3
6	HR	30 HR	IDF A	ASW1 Fa0/4–5 · ASW2 Fa0/4–6
7	Loans	70 LOANS	IDF B	ASW3 Fa0/1–4 · ASW4 Fa0/1–3
8	IT	40 IT	IDF A + IDF B	ASW1 Fa0/6–9 · ASW2 Fa0/7–9 · ASW3 Fa0/5–6 · ASW4 Fa0/4–6
9	Security Office	80 SECURITY	IDF C	VLAN 80 ports shared with room 14: ASW5 Fa0/10–11, Fa0/14–15 · ASW6 Fa0/6–8
10	Staff Room / Break	— (none assigned)	—	Open item V9
11	Network Room	— (none assigned)	—	Open item V7: purpose vs MDF in Room 12
12	Server Room · MDF	90 STORAGE_SERVERS	IDF C	MDF: L3SW1, L3SW2, Router0, edge · VLAN 90 ports shared with room 13: ASW5 Fa0/12–13 · ASW6 Fa0/9–11
13	Storage / Records	90 STORAGE_SERVERS	IDF C	See room 12
14	Security Control Room	80 SECURITY	IDF C	See room 9
15	ATM Lobby	— (none assigned)	—	Open item V9: ATMs need their own decision
—	Guest Wi-Fi (APs)	100 GUEST	IDF A + IDF C	ASW2 Fa0/10–12 · ASW6 Fa0/12–13

2.3 MDF and IDF racks

Rack	Location on the plan	Switches	Dist ports (L3SW1 + L3SW2)	VLANs served
MDF	Room 12 (Server Room)	L3SW1, L3SW2, Router0, edge	—	Routed core; Po1 Gi1/0/23–24
IDF A	Left side, beside Finance (5)	ASW1, ASW2 (PoE)	Gi1/0/3, Gi1/0/4	20, 30, 40, 60, 100
IDF B	Beside Loans (7), near the MDF	ASW3, ASW4 (PoE)	Gi1/0/5, Gi1/0/6	40, 70
IDF C	Top corridor, near Security Office (9)	ASW5, ASW6 (PoE)	Gi1/0/7, Gi1/0/8	10, 50, 80, 90, 100

Every IDF is dual-homed: Gi0/1 of each ASW goes to L3SW1 and Gi0/2 to L3SW2 in the MDF. High-security rooms (11 Network, 12 Server, 14 Security Control) are access-controlled.

Verify — Open items, not decisions: **V7** Room 11 is labelled Network Room, but the MDF is drawn in Room 12 — define Room 11's purpose. **V8** About 15 APs are drawn (one per room); earlier files had 12 APs on ASW7 — confirm the AP count and feeding ports. **V9** Rooms 1 Public Lobby (suggest Guest 100), 10 Staff Room and 15 ATM Lobby (ATMs need their own decision) have no VLAN.

3 Logical Topology

3.1 Topology diagram

Bank Main Branch — Logical Topology (Access → Distribution → Edge → WAN → Head-Quarters)

Three-tier campus · 802.1Q trunks native 99 · HSRP VIP .1 · Po1 L3 LACP · static routing hop-by-hop to HQ 172.16.10.0/24 · per PT canvas 2026-09-30

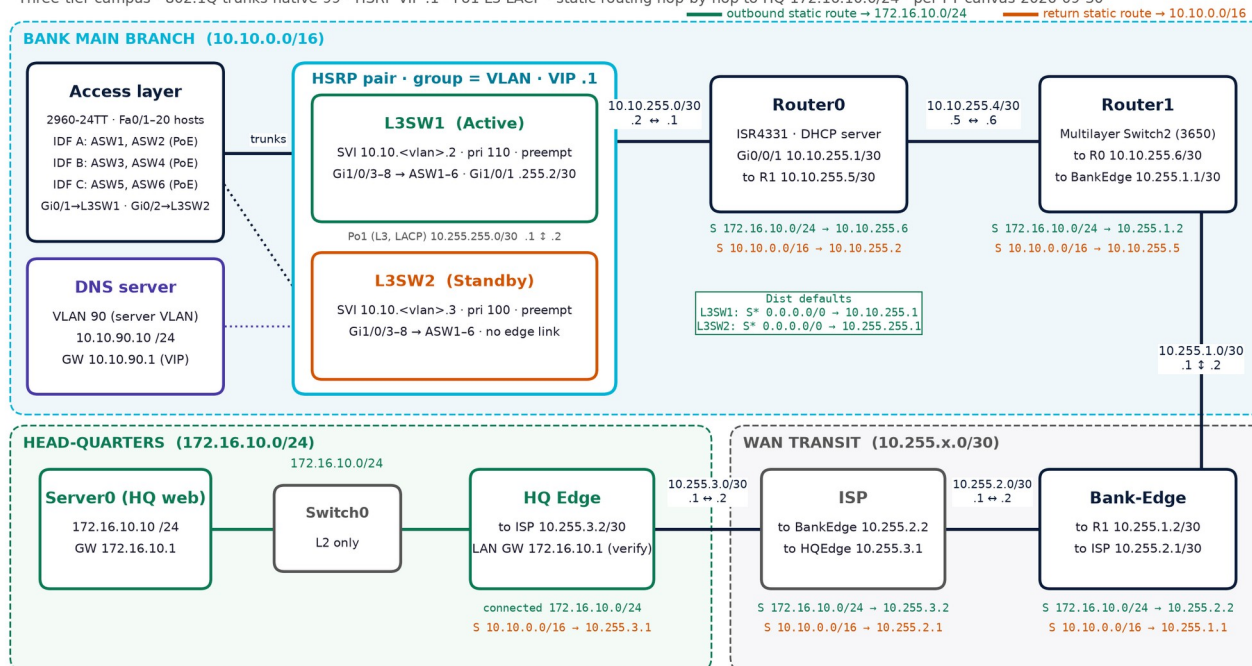


Figure 2 — Logical topology: access (IDF A/B/C) → HSRP distribution pair → Router0 → Router1 (Multilayer Switch2) → Bank-Edge → ISP → HQ Edge → Server0, with the static routes on each hop.

3.2 Tiers and traffic path

Tier	Devices	Role
Access	ASW1-ASW6 (2960-24TT)	End devices on Fa0/1-20; dual trunks Gi0/1 → L3SW1, Gi0/2 → L3SW2 (native 99)
Distribution	L3SW1, L3SW2 (3650-24PS)	SVI gateways + HSRP, DHCP relay, inter-VLAN routing, Po1 routed interconnect, default routes
Edge	Router0 (ISR4331), Router1 = Multilayer Switch2 (3650)	Router0: DHCP server + edge 10.10.255.1. Router1: routed transit toward the WAN
WAN / HQ	Bank-Edge, ISP, HQ Edge (ISR4331), Switch0, Server0	Static routing to HQ LAN 172.16.10.0/24

Branch → HQ: host → VIP (Active L3SW1) → Router0 10.10.255.1 → Router1 10.10.255.6 → Bank-Edge 10.255.1.2 → ISP 10.255.2.2 → HQ Edge 10.255.3.2 → Server0 172.16.10.10. The return path is the same hops in reverse, each using 10.10.0.0/16. When L3SW2 is Active, outside traffic goes L3SW2 → Po1 → L3SW1 → Router0.

Figure 3 — Packet Tracer logical workspace, 30 September 2026: bank campus, WAN chain and Head-Quarters. Rotated 90°: turn the page clockwise to read.



Canvas element	Meaning	Canvas element	Meaning
Colour legend (top left)	VLAN colour scheme; coloured zones group the end devices by VLAN (Section 5.1)	IDF A / B / C boxes	ASW1-2, ASW3-4, ASW5-6; ASW2/4/6 labelled PoE
Po1 LACP 10.255.255.0/30	Routed EtherChannel L3SW1 .1 ↔ L3SW2 .2 (ellipse)	"SVI configured" tags	SVIs present on L3SW1 and L3SW2
/30 labels (top)	10.10.255.0/30 R0-L3SW1 · 10.10.255.4/30 R0-MLS2 (.5/.6) · 10.255.1/2/3.0/30 WAN	Head Quarters zone	HQ Edge, Switch0, Server0 — 172.16.10.0/24, Server0 172.16.10.10
Solid / dashed links	Solid = copper straight-through · dashed = copper cross-over	"DHCP here"	Router0 is the DHCP server

Note — The legend entry **VLAN 999 NetworkManagement (10.10.99.0/24)** is not part of the bank design (it comes from a separate campus plan) and is left out of every table. VLAN 99 is the trunk native VLAN with no subnet.

5 VLAN and IP Addressing Plan

5.1 VLANs, SVIs and HSRP

VLAN	Name	Subnet	VIP (HSRP)	L3SW1 (.2)	L3SW2 (.3)	Colour
10	MANAGEMENT	10.10.10.0/24	10.10.10.1	10.10.10.2	10.10.10.3	Dark green
20	FINANCE	10.10.20.0/24	10.10.20.1	10.10.20.2	10.10.20.3	Pink
30	HR	10.10.30.0/24	10.10.30.1	10.10.30.2	10.10.30.3	Red
40	IT	10.10.40.0/24	10.10.40.1	10.10.40.2	10.10.40.3	Lavender
50	TELLERS	10.10.50.0/24	10.10.50.1	10.10.50.2	10.10.50.3	Yellow
60	CUSTOMER_SERVICE	10.10.60.0/24	10.10.60.1	10.10.60.2	10.10.60.3	Green
70	LOANS	10.10.70.0/24	10.10.70.1	10.10.70.2	10.10.70.3	Light blue
80	SECURITY	10.10.80.0/24	10.10.80.1	10.10.80.2	10.10.80.3	Peach
90	STORAGE_SERVERS	10.10.90.0/24	10.10.90.1	10.10.90.2	10.10.90.3	Brown
100	GUEST	10.10.100.0/24	10.10.100.1	10.10.100.2	10.10.100.3	Orange
99	NATIVE	— (trunk native only)	—	—	—	—

Names follow Nondumiso's VLAN database. The L3 & L7 report uses short names for the same IDs (60 CustomerSVC, 90 Servers, 100 Wireless = GUEST). Colours are the canvas legend of 30 Sep 2026.

5.2 Point-to-point links and HQ LAN

Link	Subnet	Side A	Side B
Router0 ↔ L3SW1 (edge)	10.10.255.0/30	Router0 Gi0/0/1 10.10.255.1	L3SW1 Gi1/0/1 10.10.255.2
Router0 ↔ Router1 (Multilayer Switch2)	10.10.255.4/30	Router0 10.10.255.5	Router1 10.10.255.6
Router1 ↔ Bank-Edge	10.255.1.0/30	Router1 10.255.1.1	Bank-Edge 10.255.1.2
Bank-Edge ↔ ISP	10.255.2.0/30	Bank-Edge 10.255.2.1	ISP 10.255.2.2
ISP ↔ HQ Edge	10.255.3.0/30	ISP 10.255.3.1	HQ Edge 10.255.3.2
L3SW1 ↔ L3SW2 (Po1, L3 LACP)	10.255.255.0/30	L3SW1 Po1 10.255.255.1	L3SW2 Po1 10.255.255.2
HQ LAN (Switch0)	172.16.10.0/24	HQ Edge 172.16.10.1 (VERIFY)	Server0 172.16.10.10

5.3 Service addresses

Service	Address	Where	Notes
DHCP server	10.10.255.1	Router0 Gi0/0/1	Helper target on every Dist SVI
Default gateways	10.10.<vlan>.1	HSRP VIP on L3SW1/L3SW2	DHCP default-router; hosts never use .2/.3
DNS (primary)	10.10.90.10	Server-PT, VLAN 90	Static; GW 10.10.90.1; inside the VLAN 90 exclusion
DNS (secondary)	8.8.8.8	Public	No internet path in the lab
HQ server (Server0)	172.16.10.10	HQ LAN behind HQ Edge	HTTP; target of the HQ tests (T11–T14)
DHCP exclusions	.1–.10 per VLAN	Router0	VLAN 90: .1–.49 for static servers

6 Layer 1 — Devices, Cabling and Port Map

6.1 Devices and models

Device (canvas name)	PT model	Role	Location
Router0	ISR4331	Campus edge · DHCP server	MDF, Room 12
Router1 ("Multilayer Switch2")	3650-24PS	Routed transit hop Router0 ↔ Bank-Edge (ip routing, routed ports)	MDF, Room 12
L3SW1 / L3SW2	3650-24PS	Distribution pair · SVIs · HSRP Active / Standby · Po1	MDF, Room 12
ASW1, ASW3, ASW5	2960-24TT	Access · IDF A / B / C	IDF A / B / C racks
ASW2, ASW4, ASW6 (PoE)	2960-24TT	Access · IDF A / B / C · feed the access points	IDF A / B / C racks
Bank-Edge	ISR4331	Branch WAN edge toward the ISP	Branch
ISP	ISR4331	Provider transit (lab)	Provider
HQ Edge	ISR4331	Head-Quarters gateway	Head-Quarters
Switch0	2960-24TT	HQ LAN switch (L2 only)	Head-Quarters
Server0	Server-PT	HQ web server 172.16.10.10	Head-Quarters
DNS server	Server-PT	Branch DNS 10.10.90.10 (VLAN 90)	Branch
End devices	PC-PT, Laptop-PT, AccessPoint-PT	Users and APs in the VLAN zones of the canvas	Floor plan rooms

Note — “PoE” on ASW2/4/6 is a design label. The PT 2960-24TT has no PoE ports; a physical build needs a PoE model for the access points. ASW7 from earlier files is not on the 30 Sep canvas.

6.2 Cabling

Link type	Cable (as drawn)	Links
Router ↔ switch	Copper straight-through (solid)	Router0–L3SW1 · Router0–Router1 (MLS2) · Router1–Bank-Edge · HQ Edge–Switch0
Switch ↔ switch	Copper cross-over (dashed)	ASW Gi0/1–2 ↔ Dist Gi1/0/3–8 (12 trunks) · Po1 members Gi1/0/23–24
Router ↔ router	Copper cross-over (dashed)	Bank-Edge–ISP · ISP–HQ Edge
Switch ↔ end device	Copper straight-through	PCs, laptops, APs, servers on Fa0/1–20
MDF ↔ IDF racks	Fibre backbone	Room 12 → IDF A, IDF B, IDF C (physical design; not modelled in the logical view)

6.3 Port map

Device / port	Connects to	Purpose / addressing
Router0 Gi0/0/1	L3SW1 Gi1/0/1	Edge 10.10.255.0/30 (.1 ↔ .2)
Router0 Gi0/0/0 (VERIFY)	Router1 port (VERIFY)	10.10.255.4/30 (.5 ↔ .6)
Router1 port (VERIFY)	Bank-Edge Gi0/0/0 (VERIFY)	10.255.1.0/30 (.1 ↔ .2)
Bank-Edge Gi0/0/1 (VERIFY)	ISP Gi0/0/0 (VERIFY)	10.255.2.0/30 (.1 ↔ .2)
ISP Gi0/0/1 (VERIFY)	HQ Edge Gi0/0/0 (VERIFY)	10.255.3.0/30 (.1 ↔ .2)
HQ Edge Gi0/0/1 (VERIFY)	Switch0 → Server0	HQ LAN 172.16.10.0/24
L3SW1 Gi1/0/23–24	L3SW2 Gi1/0/23–24	Po1, LACP active, routed 10.255.255.0/30
L3SW1 Gi1/0/3 ... Gi1/0/8	ASW1 ... ASW6 Gi0/1	802.1Q trunks, native 99
L3SW2 Gi1/0/3 ... Gi1/0/8	ASW1 ... ASW6 Gi0/2	802.1Q trunks, native 99 (redundant uplinks)
ASW1–ASW6 Fa0/1–20	End devices	Access VLANs per Section 7.2 · PortFast + BPDU Guard

7 Layer 2 — Switching (Nondumiso Mbuyazi)

Condensed from *Bank Network — Layer 2 & Layer 3 Configuration Commands* (Nondumiso Mbuyazi, Rev 1.0). Her script is the source of truth for L2 values. Full per-switch configurations are in Appendix A.4–A.5.

7.1 VLAN database

All switches: L3SW1, L3SW2, ASW1-ASW6	(continued)
enable configure terminal vlan 10 name MANAGEMENT vlan 20 name FINANCE vlan 30 name HR vlan 40 name IT vlan 50 name TELLERS vlan 60 name CUSTOMER_SERVICE	vlan 70 name LOANS vlan 80 name SECURITY vlan 90 name STORAGE_SERVERS vlan 99 name NATIVE vlan 100 name GUEST end write memory

7.2 Access ports

Switch	IDF	Port ranges → VLAN (all switchport mode access)
ASW1	IDF A	Fa0/1–3 → 20, Fa0/4–5 → 30, Fa0/6–9 → 40, Fa0/15–17 → 60
ASW2 (PoE)	IDF A	Fa0/1–3 → 20, Fa0/4–6 → 30, Fa0/7–9 → 40, Fa0/10–12 → 100, Fa0/13–15 → 60
ASW3	IDF B	Fa0/1–4 → 70, Fa0/5–6 → 40
ASW4 (PoE)	IDF B	Fa0/1–3 → 70, Fa0/4–6 → 40
ASW5	IDF C	Fa0/1–5 → 50, Fa0/6–9 → 10, Fa0/10–11 → 80, Fa0/12–13 → 90, Fa0/14–15 → 80
ASW6 (PoE)	IDF C	Fa0/1–3 → 10, Fa0/4–5 → 50, Fa0/6–8 → 80, Fa0/9–11 → 90, Fa0/12–13 → 100

- End-device ports Fa0/1–20 on every ASW get `spanning-tree portfast` and `spanning-tree bpduguard enable` (one `interface range fa0/1 - 20`). Never on trunks: ASW Gi0/1–2, Dist Gi1/0/3–8, Po1.
- Differences from the older port table (Bank_Main_Branch_Port_Table.csv) are listed in her command book; her ranges are used here.

7.3 Trunks

Verify — Always extend a trunk with `switchport trunk allowed vlan add <id>`. Without **add** the command replaces the list and every other VLAN drops. Native VLAN and allowed list must match on both ends.

ASW1-ASW6 uplinks (Gi0/1 -> L3SW1, Gi0/2 -> L3SW2)
! ASW1-ASW6: Gi0/1 + Gi0/2 = redundant uplinks ! Gi0/1 -> L3SW1, Gi0/2 -> L3SW2 interface range gi0/1 - 2 description UPLINK to L3SW1 / L3SW2 switchport mode trunk switchport trunk native vlan 99 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100 switchport nonegotiate no shutdown ! end write memory

L3SW1 + L3SW2 downlinks Gi1/0/3-8 (identical on both)

```
! Never Gi1/0/1 (routed edge to Router0)
! Gi1/0/3 ASW1 | /4 ASW2 | /5 ASW3 | /6 ASW4 | /7 ASW5 | /8 ASW6
interface range gigabitEthernet1/0/3 - 8
  description DOWNLINK to ASW1-ASW6
! skip the next line if the platform rejects it (dot1q only)
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100
  switchport nonegotiate
  no shutdown
!
end
write memory
```

7.4 PortFast, BPDU Guard and L3 EtherChannel

Access switches - edge ports

```
! Access switches ASW1-ASW6
! (in every ASW block, Appendix A.5)
interface range fa0/1 - 20
  spanning-tree portfast
  spanning-tree bpduguard enable
!
! Fa0/1-20 = end-device ports only.
! NEVER on trunks / uplinks:
!   ASW Gi0/1-2, Dist Gi1/0/3-8, Po1
!
! Recover a port that BPDU Guard err-disabled
! (remove the rogue switch first):
interface fastEthernet0/5
  shutdown
  no shutdown
```

L3SW1 - Po1 (L3SW2 identical, .2)

```
! L3SW1 - Po1 routed link to L3SW2
enable
configure terminal
interface range gigabitEthernet1/0/23 - 24
  description Po1 member to L3SW2
  no switchport
  channel-group 1 mode active
  no shutdown
!
interface port-channel 1
  description L3 EtherChannel to L3SW2
  no switchport
  ip address 10.255.255.1 255.255.255.252
  no shutdown
!
end
write memory
```

8 Layer 3 — HSRP, DHCP, Routing, DNS (Innocent Nkosiye Mutha)

Condensed from the *Bank Main Branch — Layer 3 & Layer 7 Configuration Report* (Rev 1.0, HQ subnet updated to 172.16.10.0/24). Full configurations are in Appendix A.1–A.4.

8.1 SVIs and HSRP

Parameter	Value	Why
Group / VIP	Group = VLAN ID · VIP 10.10.<vlan>.1	<code>show standby brief</code> maps 1:1 to VLANs; VIP = DHCP default-router
L3SW1	.2 · priority 110 · preempt	Deterministic Active; takes the role back after recovery
L3SW2	.3 · priority 100 · preempt	Standby; Active only if the L3SW1 group fails
Timers / version	PT default v1, hello 3 s / hold 10 s	Groups 10–100 fit the v1 range
L2 dependency	Hellos cross the ASW trunks	Po1 is routed, so VLAN adjacency between the Dist runs through the dual-homed ASWs
Rule (TKT-2109)	No SVI holds .1 as its real IP	A real .1 on an SVI fights the VIP (ARP/gateway conflict)

L3SW1 - Active (VLAN 20 pattern)	L3SW2 - Standby (VLAN 20 pattern)
<pre>interface Vlan20 description FINANCE - HSRP Active ip address 10.10.20.2 255.255.255.0 ip helper-address 10.10.255.1 standby 20 ip 10.10.20.1 standby 20 priority 110 standby 20 preempt no shutdown !</pre>	<pre>interface Vlan20 description FINANCE - HSRP Standby ip address 10.10.20.3 255.255.255.0 ip helper-address 10.10.255.1 standby 20 ip 10.10.20.1 standby 20 priority 100 standby 20 preempt no shutdown !</pre>

- **Failover:** if the L3SW1 group fails, L3SW2 becomes Active after the hold time and hosts keep the same gateway; preempt returns Active to L3SW1.
- **Known limit:** a whole-L3SW1 failure keeps the VIP alive, but Router0's only edge link is on L3SW1, so DHCP and HQ access stop.

8.2 DHCP

- Router0 is the **only** DHCP server; the Dist switches relay with `ip helper-address 10.10.255.1` on **both** L3SW1 and L3SW2.
- The relay sets giaddr to the SVI address (.2/.3); Router0 picks the pool whose network contains giaddr and replies via 10.10.0.0/16.
- Exclusions .1–.10 (VLAN 90 .1–.49). No pool for VLAN 99.

VLAN	Pool	network	default-router	dns-server	Leases
10	VLAN10_MGMT	10.10.10.0/24	10.10.10.1	10.10.90.10 8.8.8.8	.11–.254
20	VLAN20_FINANCE	10.10.20.0/24	10.10.20.1	10.10.90.10 8.8.8.8	.11–.254
30	VLAN30_HR	10.10.30.0/24	10.10.30.1	10.10.90.10 8.8.8.8	.11–.254
40	VLAN40_IT	10.10.40.0/24	10.10.40.1	10.10.90.10 8.8.8.8	.11–.254
50	VLAN50_TELLERS	10.10.50.0/24	10.10.50.1	10.10.90.10 8.8.8.8	.11–.254
60	VLAN60_CUSTSVC	10.10.60.0/24	10.10.60.1	10.10.90.10 8.8.8.8	.11–.254
70	VLAN70_LOANS	10.10.70.0/24	10.10.70.1	10.10.90.10 8.8.8.8	.11–.254
80	VLAN80_SECURITY	10.10.80.0/24	10.10.80.1	10.10.90.10 8.8.8.8	.11–.254
90	VLAN90_SERVERS	10.10.90.0/24	10.10.90.1	10.10.90.10 8.8.8.8	.50–.254
100	VLAN100_WIFI	10.10.100.0/24	10.10.100.1	10.10.90.10 8.8.8.8	.11–.254

Router0 - pool pattern	Dist - relay pattern
<pre>! Router0 - one pool per user VLAN (VLAN 20 shown) ip dhcp excluded-address 10.10.20.1 10.10.20.10 ip dhcp pool VLAN20_FINANCE network 10.10.20.0 255.255.255.0 default-router 10.10.20.1 dns-server 10.10.90.10 8.8.8.8 ! ! Full set: Appendix A.1</pre>	<pre>! L3SW1 AND L3SW2 - every user SVI interface Vlan20 ip helper-address 10.10.255.1 ! ! Target = Router0 edge IP (DHCP server). ! Never a VIP, an SVI or 10.255.255.x. ! Repeat for Vlan10 ... Vlan100 (not 99).</pre>

8.3 Static routing (internal and to HQ)

- `ip routing` on both Dist switches makes every SVI a connected route; inter-VLAN traffic is routed on the Active switch.
- L3SW1 is connected to Router0, so its default uses **10.10.255.1**. L3SW2 has no interface in 10.10.255.0/30, so its default uses L3SW1's Po1 address **10.255.255.1**.
- A next hop must be on a connected subnet and never an address the device owns, or IOS returns `%Invalid next hop address (it's this router)`.

Device	Destination	Next hop	Purpose
L3SW1	0.0.0.0/0	10.10.255.1 (Router0)	Outside / HQ
L3SW2	0.0.0.0/0	10.255.255.1 (L3SW1 Po1)	Outside / HQ + DHCP relay
Router0	10.10.0.0/16 · 10.255.255.0/30	10.10.255.2 (L3SW1)	Return to the branch and Po1
Router0	172.16.10.0/24	10.10.255.6 (Router1)	Outbound to HQ
Router1 (MLS2)	172.16.10.0/24 · 10.10.0.0/16	10.255.1.2 · 10.10.255.5	Outbound · return
Bank-Edge	172.16.10.0/24 · 10.10.0.0/16	10.255.2.2 · 10.255.1.1	Outbound · return
ISP	172.16.10.0/24 · 10.10.0.0/16	10.255.3.2 · 10.255.2.1	Outbound · return
HQ Edge	10.10.0.0/16 · 0.0.0.0/0	10.255.3.1 · 10.255.3.1	Return · default

Static routes - branch	Static routes - WAN and HQ
<pre>! L3SW1 (owns the edge link) ip route 0.0.0.0 0.0.0.0 10.10.255.1 ! L3SW2 (no edge link: via L3SW1 Po1) ip route 0.0.0.0 0.0.0.0 10.255.255.1 ! Router0 ip route 10.10.0.0 255.255.0.0 10.10.255.2 ip route 172.16.10.0 255.255.255.0 10.10.255.6 ip route 10.255.255.0 255.255.255.252 10.10.255.2 ! Router1 (Multilayer Switch2) ip route 172.16.10.0 255.255.255.0 10.255.1.2 ip route 10.10.0.0 255.255.0.0 10.10.255.5</pre>	<pre>! Bank-Edge ip route 172.16.10.0 255.255.255.0 10.255.2.2 ip route 10.10.0.0 255.255.0.0 10.255.1.1 ! ISP ip route 172.16.10.0 255.255.255.0 10.255.3.2 ip route 10.10.0.0 255.255.0.0 10.255.2.1 ! HQ Edge ip route 10.10.0.0 255.255.0.0 10.255.3.1 ip route 0.0.0.0 0.0.0.0 10.255.3.1 ! ! Every hop: outbound 172.16.10.0/24 + ! return 10.10.0.0/16 (both directions)</pre>

Note — Router1 is the 3650 “Multilayer Switch2” on the canvas, so its block uses `ip routing` and no `switchport` routed ports (Appendix A.2). The L3 & L7 report still shows an ISR-style Router1 block.

8.4 DNS

Name	Type	Address	Status
router0	A	10.10.255.1	Live (2026-09-16 extract)
inno@bank.com	A	10.10.40.53	Live — '@' is not valid in a hostname; rename (e.g. inno.bank.com)
hq-web.bank.com	A	172.16.10.10	Proposed — add once the HQ path (TKT-2114/2115) is green

- DNS server Server-PT **10.10.90.10**, VLAN 90, static, gateway 10.10.90.1 (VIP) so it survives HSRP failover.
- Every pool hands out `dns-server 10.10.90.10 8.8.8.8`. 8.8.8.8 never answers in the lab (no internet path).

9 Testing and Verification

Verify — Results are **not** filled in by this document. Run each test on the current .pkt, record the outcome and a screenshot reference in the last column, and treat any difference from the expected result as a finding.

9.1 Test plan

#	Area	Test / command	Expected result	Result / screenshot
T01	Trunks	show interfaces trunk (ASW1-6, L3SW1/2)	ASW Gi0/1-2 and Dist Gi1/0/3-8 trunking, native 99, same allowed list both ends	
T02	Po1	show etherchannel summary (L3SW1/2)	Po1(RU), LACP, Gi1/0/23 and Gi1/0/24 (P); ping 10.255.255.2 from L3SW1 replies	
T03	VLANs / access	show vlan brief (all switches)	VLANs 10-100 + 99 with their names; each access range in its VLAN	
T04	HSRP state	show standby brief (L3SW1, L3SW2)	10 groups; L3SW1 Active pri 110 P; L3SW2 Standby pri 100 P; VIP .1; no Vlan99	
T05	DHCP lease	Finance PC → IP Configuration DHCP	10.10.20.11-.254, /24, GW 10.10.20.1, DNS 10.10.90.10	
T06	DHCP leases (all VLANs)	One host per VLAN 10-100 → DHCP; Router0 show ip dhcp binding	A binding in every pool; none inside an excluded range	
T07	Gateway	Finance PC ping 10.10.20.1	Replies from the VIP	
T08	Inter-VLAN	Finance PC ping an HR host (VLAN 30)	Replies, routed on the Active Dist	
T09	Edge	Finance PC ping 10.10.255.1	Replies from Router0 (default route + return route)	
T10	Routes	show ip route on each hop	Entries exactly as Section 8.3; next hops on connected /30s	
T11	HQ reachability	Finance PC ping 172.16.10.10	Replies from Server0	
T12	Path	Finance PC tracert 172.16.10.10	10.10.20.2 → 10.10.255.1 → 10.10.255.6 → 10.255.1.2 → 10.255.2.2 → 10.255.3.2 → 172.16.10.10	
T13	DNS	PC nslookup router0	Server 10.10.90.10 answers 10.10.255.1	
T14	HQ web	PC Web Browser http://172.16.10.10	Server0 page loads	
T15	HSRP failover	PC ping -t 10.10.20.1 + ping -t 172.16.10.10; L3SW1 interface Vlan20 → shutdown	L3SW2 Active for group 20 after ≈10 s; VIP pings resume; HQ pings resume via L3SW2 → Po1 → L3SW1 (TKT-2115)	
T16	DHCP in failover	During T15: renew a Finance PC	New lease via the L3SW2 helper (TKT-2111)	
T17	Preempt	L3SW1 interface Vlan20 → no shutdown	L3SW1 back to Active (110, preempt); L3SW2 Standby; pings continue	
T18	BPDU Guard	Connect a switch to an ASW Fa0/x port	Port goes err-disabled; recover with shutdown / no shutdown	

10 Troubleshooting Log

Sources: drill board waves 1–12 (15–30 Sep 2026), case study, hq-fix note, live extract 2026-09-16. Status as last recorded; close a ticket only with the proof listed.

10.1 Resolved and diagnosed

INC-1042	Resolved • DHCP / gateway
Problem	DHCP / default-gateway fault (resolved).
Cause	Not recorded in the project files — VERIFY : add from the ticket notes.
Fix	Not recorded in the project files — VERIFY .
Proof	To be attached.

INC-2101	Resolved • Finance VLAN 20 — HSRP Init
Problem	Finance PCs Limited/No connectivity and banking apps down; HR (VLAN 30) next door fine. HSRP group 20 in Init.
Cause	Duplicate HSRP priority 110 on both Dist switches (no deterministic Active) and missing trunks, so the VLAN 20 hellos did not reach the peer.
Fix	L3SW1 .2 priority 110 + preempt (Active), L3SW2 .3 priority 100 + preempt (Standby), VIP 10.10.20.1; trunks restored (native 99, VLAN 20 allowed).
Proof	<code>show standby brief</code> group 20 Active/Standby on both; Finance PC pings 10.10.20.1; HR still healthy. Screenshots cv-screenshots/03-L3SW1, 04-L3SW2.

TKT-2104	Resolved • DHCP pools + helpers
Problem	VIP pinged, but Finance (PC19) and Tellers (VLAN 50) stayed on APIPA; Discover never became Offer.
Cause	No relay path to the DHCP server: helpers missing on the Dist SVIs and the Dist had to reach Router0 over the edge /30.
Fix	Edge /30 Router0 .1 ↔ L3SW1 .2; <code>ip helper-address 10.10.255.1</code> on the Dist SVIs; one Router0 pool per VLAN with default-router = VIP and exclusions .1–.10.
Proof	Client IP/mask/GW VIP/DNS; Router0 <code>show ip dhcp binding</code> with Finance 10.10.20.x leases (cv-screenshots/05-Router0-DHCP).

LESSON-01	Lesson learned • L3SW1 default route
Problem	Adding the Dist default route returned <code>%Invalid next hop address (it's this router)</code> .
Cause	The next hop was an address the switch owns itself (its own SVI .2/.3 or 10.10.255.2 on L3SW1).
Fix	Point at the neighbour: L3SW1 <code>ip route 0.0.0.0 0.0.0.0 10.10.255.1</code> ; L3SW2 (no edge link) <code>ip route 0.0.0.0 0.0.0.0 10.255.255.1</code> .
Proof	<code>show ip route</code> → S* 0.0.0.0/0 via 10.10.255.1, gateway of last resort 10.10.255.1 (capture pending).

TKT-2114	Diagnosed • Dist missing outside route to HQ
Problem	Bank PCs cannot reach Server0 172.16.10.10. Router0 reaches the HQ server and L3SW1 reaches 10.10.255.1, but L3SW1 and the PCs cannot reach HQ.
Cause	The Dist switches have no route outside the branch (no default / HQ route).
Fix (to apply)	L3SW1 default → 10.10.255.1, L3SW2 default → 10.255.255.1; confirm Router0 172.16.10.0/24 → 10.10.255.6 and the 10.10.0.0/16 return on every hop.
Proof	Pending — tests T11 and T12.

10.2 Open tickets

Ticket	Problem (last recorded)	Next action	Status
TKT-2109	VLAN 60: L3SW2 Vlan60 owns 10.10.60.1 as its real IP with no <code>standby 60</code> — gateway conflict for Customer Service	L3SW2 <code>ip address 10.10.60.3 255.255.255.0 + standby 60 ip 10.10.60.1</code> , priority 100, preempt; T04	Open
TKT-2105	Finance VLAN 20 HSRP failover not yet proven	Run T15–T17 and attach both <code>show standby brief</code> captures	Open
TKT-2111	Helpers mainly on L3SW1; DHCP during Active loss unproven	Helpers on all L3SW2 SVIs + L3SW2 default; T16	Open
TKT-2113	Security VLAN 80 hosts get a lease but cannot use VIP 10.10.80.1 (default-router / VIP mismatch)	Pool VLAN80_SECURITY <code>default-router 10.10.80.1</code> ; check group 80; T05 on VLAN 80	Open
TKT-2115	HQ reachable only while L3SW1 is Active; L3SW2 has no working outside route	L3SW2 <code>ip route 0.0.0.0 0.0.0.0 10.255.255.1</code> ; T15 HQ ping	Open

10.3 Items to verify in the live file

#	Item	Why
V1	HQ Edge LAN gateway 172.16.10.1 and Server0 gateway	Not labelled on the canvas; set to match
V2	Port IDs on Router0 Gi0/0/0, Router1 (MLS2), Bank-Edge, ISP, HQ Edge	No extract lists them — <code>show ip interface brief / CDP</code>
V3	Extra cables on the canvas: L3SW1 (.9) ↔ Multilayer Switch2 (.10), Multilayer Switch2 ↔ L3SW2, Router0 ↔ L3SW2	Not in the design lock; document their addressing or remove them
V4	Po1 member lights shown amber on the 30 Sep capture	Confirm Po1(RU) with <code>show etherchannel summary (T02)</code>
V5	Router1 hostname on the 3650 (canvas label “Multilayer Switch2”)	Configs use <code>hostname Router1</code>
V6	INC-1042 cause, fix and proof	Not in the project files
V7	Room 11 “Network Room” vs the MDF drawn in Room 12 (Server Room)	Room 11 has no defined purpose or VLAN
V8	≈15 APs drawn on the floor plan (one per room) vs 12 APs on ASW7 in earlier files	Confirm AP count and which ASW ports feed them
V9	No VLAN for rooms 1 Public Lobby, 10 Staff Room, 15 ATM Lobby	Suggestion: Lobby → Guest 100; ATMs need their own decision

Appendix A — Running Configurations (Target State)

Paste-ready per device, in configuration order. Comments are on their own lines (IOS rejects inline ! comments). Lines marked ! VERIFY use interface IDs that are not in any extract.

A.1 Router0

Router0 - interfaces and static routes

```
! ===== Router0 (ISR4331) - campus edge + DHCP server =====
enable
configure terminal
hostname Router0
!
interface GigabitEthernet0/0/1
description EDGE to L3SW1 Gi1/0/1 (10.10.255.0/30)
ip address 10.10.255.1 255.255.255.252
no shutdown
!
! VERIFY port ID (canvas: Router0 - Multilayer Switch2)
interface GigabitEthernet0/0/0
description to Router1 (10.10.255.4/30)
ip address 10.10.255.5 255.255.255.252
no shutdown
!
! Return to all bank VLANs via L3SW1
ip route 10.10.0.0 255.255.0.0 10.10.255.2
! Outbound to HQ LAN via Router1
ip route 172.16.10.0 255.255.255.0 10.10.255.6
! Return for L3SW2-sourced traffic (Po1 is outside 10.10.0.0/16)
ip route 10.255.255.0 255.255.255.252 10.10.255.2
!
! DHCP server: see next box
end
write memory
```

Router0 - DHCP (1/2)

```
! ===== Router0 - DHCP (1/2) =====
service dhcp
! Exclusions: .1 VIP, .2/.3 Dist, infra to .10
! VLAN 90: .1-.49 static servers
ip dhcp excluded-address 10.10.10.1 10.10.10.10
ip dhcp excluded-address 10.10.20.1 10.10.20.10
ip dhcp excluded-address 10.10.30.1 10.10.30.10
ip dhcp excluded-address 10.10.40.1 10.10.40.10
ip dhcp excluded-address 10.10.50.1 10.10.50.10
ip dhcp excluded-address 10.10.60.1 10.10.60.10
ip dhcp excluded-address 10.10.70.1 10.10.70.10
ip dhcp excluded-address 10.10.80.1 10.10.80.10
ip dhcp excluded-address 10.10.90.1 10.10.90.49
ip dhcp excluded-address 10.10.100.1 10.10.100.10
!
ip dhcp pool VLAN10_MGMT
network 10.10.10.0 255.255.255.0
default-router 10.10.10.1
dns-server 10.10.90.10 8.8.8.8
!
ip dhcp pool VLAN20_FINANCE
network 10.10.20.0 255.255.255.0
default-router 10.10.20.1
dns-server 10.10.90.10 8.8.8.8
!
ip dhcp pool VLAN30_HR
network 10.10.30.0 255.255.255.0
default-router 10.10.30.1
dns-server 10.10.90.10 8.8.8.8
!
ip dhcp pool VLAN40_IT
network 10.10.40.0 255.255.255.0
default-router 10.10.40.1
dns-server 10.10.90.10 8.8.8.8
!
```

Router0 - DHCP (2/2)

```
! ===== Router0 - DHCP (2/2) =====
ip dhcp pool VLAN50_TELLERS
network 10.10.50.0 255.255.255.0
default-router 10.10.50.1
dns-server 10.10.90.10 8.8.8.8
!
ip dhcp pool VLAN60_CUSTSVC
network 10.10.60.0 255.255.255.0
default-router 10.10.60.1
dns-server 10.10.90.10 8.8.8.8
!
ip dhcp pool VLAN70_LOANS
network 10.10.70.0 255.255.255.0
default-router 10.10.70.1
dns-server 10.10.90.10 8.8.8.8
!
ip dhcp pool VLAN80_SECURITY
network 10.10.80.0 255.255.255.0
default-router 10.10.80.1
dns-server 10.10.90.10 8.8.8.8
!
ip dhcp pool VLAN90_SERVERS
network 10.10.90.0 255.255.255.0
default-router 10.10.90.1
dns-server 10.10.90.10 8.8.8.8
!
ip dhcp pool VLAN100_WIFI
network 10.10.100.0 255.255.255.0
default-router 10.10.100.1
dns-server 10.10.90.10 8.8.8.8
!
! No pool for VLAN 99 (native only)
end
write memory
```

A.2 Router1 (Multilayer Switch2) and Bank-Edge

Router1 (3650 routed hop)	Bank-Edge
<pre>! ===== Router1 = "Multilayer Switch2" ===== ! 3650-24PS used as a routed hop (per canvas) enable configure terminal hostname Router1 ip routing ! ! VERIFY port ID (canvas: to Router0) interface GigabitEthernet1/0/1 description to Router0 (10.10.255.4/30) no switchport ip address 10.10.255.6 255.255.255.252 no shutdown ! ! VERIFY port ID (canvas: to Bank-Edge) interface GigabitEthernet1/0/2 description to Bank-Edge (10.255.1.0/30) no switchport ip address 10.255.1.1 255.255.255.252 no shutdown ! ip route 172.16.10.0 255.255.255.0 10.255.1.2 ip route 10.10.0.0 255.255.0.0 10.10.255.5 end write memory</pre>	<pre>! ===== Bank-Edge (ISR4331) - WAN edge ===== enable configure terminal hostname Bank-Edge ! ! VERIFY port ID interface GigabitEthernet0/0/0 description to Router1 (10.255.1.0/30) ip address 10.255.1.2 255.255.255.252 no shutdown ! ! VERIFY port ID interface GigabitEthernet0/0/1 description to ISP (10.255.2.0/30) ip address 10.255.2.1 255.255.255.252 no shutdown ! ip route 172.16.10.0 255.255.255.0 10.255.2.2 ip route 10.10.0.0 255.255.0.0 10.255.1.1 end write memory</pre>

A.3 ISP, HQ Edge and servers

ISP	HQ Edge
<pre>! ===== ISP (ISR4331) - provider transit ===== enable configure terminal hostname ISP ! ! VERIFY port ID interface GigabitEthernet0/0/0 description to Bank-Edge (10.255.2.0/30) ip address 10.255.2.2 255.255.255.252 no shutdown ! ! VERIFY port ID interface GigabitEthernet0/0/1 description to HQ-Edge (10.255.3.0/30) ip address 10.255.3.1 255.255.255.252 no shutdown ! ip route 172.16.10.0 255.255.255.0 10.255.3.2 ip route 10.10.0.0 255.255.0.0 10.255.2.1 end write memory</pre>	<pre>! ===== HQ Edge (ISR4331) - Head-Quarters ===== enable configure terminal hostname HQ-Edge ! ! VERIFY port ID interface GigabitEthernet0/0/0 description to ISP (10.255.3.0/30) ip address 10.255.3.2 255.255.255.252 no shutdown ! ! VERIFY port ID and LAN GW (not on canvas) interface GigabitEthernet0/0/1 description HQ LAN via Switch0 (172.16.10.0/24) ip address 172.16.10.1 255.255.255.0 no shutdown ! ! Return to the bank branch ip route 10.10.0.0 255.255.0.0 10.255.3.1 ! Default toward ISP (hq-fix note) ip route 0.0.0.0 0.0.0.0 10.255.3.1 end write memory</pre>

Server0, Switch0 and branch DNS server - settings
<pre>! ===== Server0 (HQ, Server-PT) - GUI ===== ! Desktop > IP Configuration (Static) ! IPv4 172.16.10.10 ! Mask 255.255.255.0 ! Gateway 172.16.10.1 (VERIFY) ! DNS 10.10.90.10 ! Services > HTTP: On ! ! ===== Switch0 (HQ, 2960-24TT) ===== ! L2 only, all ports VLAN 1 (factory default) ! ! ===== DNS server (branch, Server-PT) - GUI ===== ! Desktop > IP Configuration (Static) ! IPv4 10.10.90.10 ! Mask 255.255.255.0 ! Gateway 10.10.90.1 (HSRP VIP) ! DNS 10.10.90.10 ! Services > DNS: On (records in 8.4) ! Access port in VLAN 90; .10 is inside the ! VLAN 90 DHCP exclusion (.1-.49)</pre>

A.4 L3SW1 and L3SW2

L3SW1 (1/3)	L3SW2 (1/3)
<pre>! ===== L3SW1 (1/3) - base, VLANs, trunks, Po1 ===== enable configure terminal hostname L3SW1 ip routing ! vlan 10 name MANAGEMENT vlan 20 name FINANCE vlan 30 name HR vlan 40 name IT vlan 50 name TELLERS vlan 60 name CUSTOMER_SERVICE vlan 70 name LOANS vlan 80 name SECURITY vlan 90 name STORAGE_SERVERS vlan 99 name NATIVE vlan 100 name GUEST ! ! Gi1/0/3 ASW1 /4 ASW2 /5 ASW3 /6 ASW4 /7 ASW5 /8 ASW6 interface range gigabitEthernet1/0/3 - 8 description DOWNLINK to ASW1-ASW6 ! skip the next line if the platform rejects it (dot1q only) switchport trunk encapsulation dot1q switchport mode trunk switchport trunk native vlan 99 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100 switchport nonegotiate no shutdown ! interface range gigabitEthernet1/0/23 - 24 description Po1 member to L3SW2 no switchport channel-group 1 mode active no shutdown ! interface Port-channel1 description L3 EtherChannel to L3SW2 no switchport ip address 10.255.255.1 255.255.255.252 no shutdown ! interface GigabitEthernet1/0/1 description EDGE to Router0 Gi0/0/1 no switchport ip address 10.10.255.2 255.255.255.252 no shutdown !</pre>	<pre>! ===== L3SW2 (1/3) - base, VLANs, trunks, Po1 ===== enable configure terminal hostname L3SW2 ip routing ! vlan 10 name MANAGEMENT vlan 20 name FINANCE vlan 30 name HR vlan 40 name IT vlan 50 name TELLERS vlan 60 name CUSTOMER_SERVICE vlan 70 name LOANS vlan 80 name SECURITY vlan 90 name STORAGE_SERVERS vlan 99 name NATIVE vlan 100 name GUEST ! ! Gi1/0/3 ASW1 /4 ASW2 /5 ASW3 /6 ASW4 /7 ASW5 /8 ASW6 interface range gigabitEthernet1/0/3 - 8 description DOWNLINK to ASW1-ASW6 ! skip the next line if the platform rejects it (dot1q only) switchport trunk encapsulation dot1q switchport mode trunk switchport trunk native vlan 99 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100 switchport nonegotiate no shutdown ! interface range gigabitEthernet1/0/23 - 24 description Po1 member to L3SW1 no switchport channel-group 1 mode active no shutdown ! interface Port-channel1 description L3 EtherChannel to L3SW1 no switchport ip address 10.255.255.2 255.255.255.252 no shutdown ! ! Gi1/0/1 unused on L3SW2 (edge link is on L3SW1) !</pre>

L3SW1 (2/3)	L3SW2 (2/3)
<pre>! ===== L3SW1 (2/3) - SVIs + HSRP + helpers ===== interface Vlan10 description MANAGEMENT - HSRP Active ip address 10.10.10.2 255.255.255.0 ip helper-address 10.10.255.1 standby 10 ip 10.10.10.1 standby 10 priority 110 standby 10 preempt no shutdown ! interface Vlan20 description FINANCE - HSRP Active ip address 10.10.20.2 255.255.255.0 ip helper-address 10.10.255.1 standby 20 ip 10.10.20.1 standby 20 priority 110 standby 20 preempt no shutdown ! interface Vlan30 description HR - HSRP Active ip address 10.10.30.2 255.255.255.0 ip helper-address 10.10.255.1 standby 30 ip 10.10.30.1 standby 30 priority 110 standby 30 preempt no shutdown ! interface Vlan40 description IT - HSRP Active ip address 10.10.40.2 255.255.255.0 ip helper-address 10.10.255.1 standby 40 ip 10.10.40.1 standby 40 priority 110 standby 40 preempt no shutdown ! interface Vlan50 description TELLERS - HSRP Active ip address 10.10.50.2 255.255.255.0 ip helper-address 10.10.255.1 standby 50 ip 10.10.50.1 standby 50 priority 110 standby 50 preempt no shutdown !</pre>	<pre>! ===== L3SW2 (2/3) - SVIs + HSRP + helpers ===== interface Vlan10 description MANAGEMENT - HSRP Standby ip address 10.10.10.3 255.255.255.0 ip helper-address 10.10.255.1 standby 10 ip 10.10.10.1 standby 10 priority 100 standby 10 preempt no shutdown ! interface Vlan20 description FINANCE - HSRP Standby ip address 10.10.20.3 255.255.255.0 ip helper-address 10.10.255.1 standby 20 ip 10.10.20.1 standby 20 priority 100 standby 20 preempt no shutdown ! interface Vlan30 description HR - HSRP Standby ip address 10.10.30.3 255.255.255.0 ip helper-address 10.10.255.1 standby 30 ip 10.10.30.1 standby 30 priority 100 standby 30 preempt no shutdown ! interface Vlan40 description IT - HSRP Standby ip address 10.10.40.3 255.255.255.0 ip helper-address 10.10.255.1 standby 40 ip 10.10.40.1 standby 40 priority 100 standby 40 preempt no shutdown ! interface Vlan50 description TELLERS - HSRP Standby ip address 10.10.50.3 255.255.255.0 ip helper-address 10.10.255.1 standby 50 ip 10.10.50.1 standby 50 priority 100 standby 50 preempt no shutdown !</pre>

L3SW1 (3/3)	L3SW2 (3/3)
<pre> ! ===== L3SW1 (3/3) - SVIs + HSRP + helpers ===== interface Vlan60 description CUSTOMER_SERVICE - HSRP Active ip address 10.10.60.2 255.255.255.0 ip helper-address 10.10.255.1 standby 60 ip 10.10.60.1 standby 60 priority 110 standby 60 preempt no shutdown ! interface Vlan70 description LOANS - HSRP Active ip address 10.10.70.2 255.255.255.0 ip helper-address 10.10.255.1 standby 70 ip 10.10.70.1 standby 70 priority 110 standby 70 preempt no shutdown ! interface Vlan80 description SECURITY - HSRP Active ip address 10.10.80.2 255.255.255.0 ip helper-address 10.10.255.1 standby 80 ip 10.10.80.1 standby 80 priority 110 standby 80 preempt no shutdown ! interface Vlan90 description STORAGE_SERVERS - HSRP Active ip address 10.10.90.2 255.255.255.0 ip helper-address 10.10.255.1 standby 90 ip 10.10.90.1 standby 90 priority 110 standby 90 preempt no shutdown ! interface Vlan100 description GUEST - HSRP Active ip address 10.10.100.2 255.255.255.0 ip helper-address 10.10.255.1 standby 100 ip 10.10.100.1 standby 100 priority 110 standby 100 preempt no shutdown ! ! VLAN 99 = native only: no SVI, no HSRP, no helper ! ! Outside / HQ: default to Router0 (connected) ip route 0.0.0.0 0.0.0.0 10.10.255.1 end write memory </pre>	<pre> ! ===== L3SW2 (3/3) - SVIs + HSRP + helpers ===== interface Vlan60 description CUSTOMER_SERVICE - HSRP Standby ip address 10.10.60.3 255.255.255.0 ip helper-address 10.10.255.1 standby 60 ip 10.10.60.1 standby 60 priority 100 standby 60 preempt no shutdown ! interface Vlan70 description LOANS - HSRP Standby ip address 10.10.70.3 255.255.255.0 ip helper-address 10.10.255.1 standby 70 ip 10.10.70.1 standby 70 priority 100 standby 70 preempt no shutdown ! interface Vlan80 description SECURITY - HSRP Standby ip address 10.10.80.3 255.255.255.0 ip helper-address 10.10.255.1 standby 80 ip 10.10.80.1 standby 80 priority 100 standby 80 preempt no shutdown ! interface Vlan90 description STORAGE_SERVERS - HSRP Standby ip address 10.10.90.3 255.255.255.0 ip helper-address 10.10.255.1 standby 90 ip 10.10.90.1 standby 90 priority 100 standby 90 preempt no shutdown ! interface Vlan100 description GUEST - HSRP Standby ip address 10.10.100.3 255.255.255.0 ip helper-address 10.10.255.1 standby 100 ip 10.10.100.1 standby 100 priority 100 standby 100 preempt no shutdown ! ! VLAN 99 = native only: no SVI, no HSRP, no helper ! ! Outside / HQ + DHCP relay: default via L3SW1 Po1 ! (10.10.255.1 is not connected on L3SW2) ip route 0.0.0.0 0.0.0.0 10.255.255.1 end write memory </pre>

A.5 Access switches ASW1-ASW6

ASW1	ASW2
<pre> ! ===== ASW1 - IDF A - 2960-24TT ===== enable configure terminal hostname ASW1 ! vlan 10 name MANAGEMENT vlan 20 name FINANCE vlan 30 name HR vlan 40 name IT vlan 50 name TELLERS vlan 60 name CUSTOMER_SERVICE vlan 70 name LOANS vlan 80 name SECURITY vlan 90 name STORAGE_SERVERS vlan 99 name NATIVE vlan 100 name GUEST ! interface range fa0/1 - 3 switchport mode access switchport access vlan 20 ! interface range fa0/4 - 5 switchport mode access switchport access vlan 30 ! interface range fa0/6 - 9 switchport mode access switchport access vlan 40 ! interface range fa0/15 - 17 switchport mode access switchport access vlan 60 ! ! PortFast + BPDU Guard: end-device ports only interface range fa0/1 - 20 spanning-tree portfast spanning-tree bpduguard enable ! ! Uplinks: Gi0/1 -> L3SW1 Gi1/0/3, Gi0/2 -> L3SW2 Gi1/0/3 interface range gi0/1 - 2 description UPLINK to L3SW1 / L3SW2 switchport mode trunk switchport trunk native vlan 99 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100 switchport nonegotiate no shutdown ! end write memory </pre>	<pre> ! ===== ASW2 - IDF A (PoE) - 2960-24TT ===== enable configure terminal hostname ASW2 ! vlan 10 name MANAGEMENT vlan 20 name FINANCE vlan 30 name HR vlan 40 name IT vlan 50 name TELLERS vlan 60 name CUSTOMER_SERVICE vlan 70 name LOANS vlan 80 name SECURITY vlan 90 name STORAGE_SERVERS vlan 99 name NATIVE vlan 100 name GUEST ! interface range fa0/1 - 3 switchport mode access switchport access vlan 20 ! interface range fa0/4 - 6 switchport mode access switchport access vlan 30 ! interface range fa0/7 - 9 switchport mode access switchport access vlan 40 ! interface range fa0/10 - 12 switchport mode access switchport access vlan 100 ! interface range fa0/13 - 15 switchport mode access switchport access vlan 60 ! ! PortFast + BPDU Guard: end-device ports only interface range fa0/1 - 20 spanning-tree portfast spanning-tree bpduguard enable ! ! Uplinks: Gi0/1 -> L3SW1 Gi1/0/4, Gi0/2 -> L3SW2 Gi1/0/4 interface range gi0/1 - 2 description UPLINK to L3SW1 / L3SW2 switchport mode trunk switchport trunk native vlan 99 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100 switchport nonegotiate no shutdown ! end write memory </pre>

ASW3	ASW4
<pre> ! ===== ASW3 - IDF B - 2960-24TT ===== enable configure terminal hostname ASW3 ! vlan 10 name MANAGEMENT vlan 20 name FINANCE vlan 30 name HR vlan 40 name IT vlan 50 name TELLERS vlan 60 name CUSTOMER_SERVICE vlan 70 name LOANS vlan 80 name SECURITY vlan 90 name STORAGE_SERVERS vlan 99 name NATIVE vlan 100 name GUEST ! interface range fa0/1 - 4 switchport mode access switchport access vlan 70 ! interface range fa0/5 - 6 switchport mode access switchport access vlan 40 ! ! PortFast + BPDU Guard: end-device ports only interface range fa0/1 - 20 spanning-tree portfast spanning-tree bpduguard enable ! ! Uplinks: Gi0/1 -> L3SW1 Gi1/0/5, Gi0/2 -> L3SW2 Gi1/0/5 interface range gi0/1 - 2 description UPLINK to L3SW1 / L3SW2 switchport mode trunk switchport trunk native vlan 99 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100 switchport nonegotiate no shutdown ! end write memory </pre>	<pre> ! ===== ASW4 - IDF B (PoE) - 2960-24TT ===== enable configure terminal hostname ASW4 ! vlan 10 name MANAGEMENT vlan 20 name FINANCE vlan 30 name HR vlan 40 name IT vlan 50 name TELLERS vlan 60 name CUSTOMER_SERVICE vlan 70 name LOANS vlan 80 name SECURITY vlan 90 name STORAGE_SERVERS vlan 99 name NATIVE vlan 100 name GUEST ! interface range fa0/1 - 3 switchport mode access switchport access vlan 70 ! interface range fa0/4 - 6 switchport mode access switchport access vlan 40 ! ! PortFast + BPDU Guard: end-device ports only interface range fa0/1 - 20 spanning-tree portfast spanning-tree bpduguard enable ! ! Uplinks: Gi0/1 -> L3SW1 Gi1/0/6, Gi0/2 -> L3SW2 Gi1/0/6 interface range gi0/1 - 2 description UPLINK to L3SW1 / L3SW2 switchport mode trunk switchport trunk native vlan 99 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100 switchport nonegotiate no shutdown ! end write memory </pre>

ASW5	ASW6
<pre> ! ===== ASW5 - IDF C - 2960-24TT ===== enable configure terminal hostname ASW5 ! vlan 10 name MANAGEMENT vlan 20 name FINANCE vlan 30 name HR vlan 40 name IT vlan 50 name TELLERS vlan 60 name CUSTOMER_SERVICE vlan 70 name LOANS vlan 80 name SECURITY vlan 90 name STORAGE_SERVERS vlan 99 name NATIVE vlan 100 name GUEST ! interface range fa0/1 - 5 switchport mode access switchport access vlan 50 ! interface range fa0/6 - 9 switchport mode access switchport access vlan 10 ! interface range fa0/10 - 11 switchport mode access switchport access vlan 80 ! interface range fa0/12 - 13 switchport mode access switchport access vlan 90 ! interface range fa0/14 - 15 switchport mode access switchport access vlan 80 ! ! PortFast + BPDU Guard: end-device ports only interface range fa0/1 - 20 spanning-tree portfast spanning-tree bpduguard enable ! ! Uplinks: Gi0/1 -> L3SW1 Gi1/0/7, Gi0/2 -> L3SW2 Gi1/0/7 interface range gi0/1 - 2 description UPLINK to L3SW1 / L3SW2 switchport mode trunk switchport trunk native vlan 99 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100 switchport nonegotiate no shutdown ! end write memory </pre>	<pre> ! ===== ASW6 - IDF C (PoE) - 2960-24TT ===== enable configure terminal hostname ASW6 ! vlan 10 name MANAGEMENT vlan 20 name FINANCE vlan 30 name HR vlan 40 name IT vlan 50 name TELLERS vlan 60 name CUSTOMER_SERVICE vlan 70 name LOANS vlan 80 name SECURITY vlan 90 name STORAGE_SERVERS vlan 99 name NATIVE vlan 100 name GUEST ! interface range fa0/1 - 3 switchport mode access switchport access vlan 10 ! interface range fa0/4 - 5 switchport mode access switchport access vlan 50 ! interface range fa0/6 - 8 switchport mode access switchport access vlan 80 ! interface range fa0/9 - 11 switchport mode access switchport access vlan 90 ! interface range fa0/12 - 13 switchport mode access switchport access vlan 100 ! ! PortFast + BPDU Guard: end-device ports only interface range fa0/1 - 20 spanning-tree portfast spanning-tree bpduguard enable ! ! Uplinks: Gi0/1 -> L3SW1 Gi1/0/8, Gi0/2 -> L3SW2 Gi1/0/8 interface range gi0/1 - 2 description UPLINK to L3SW1 / L3SW2 switchport mode trunk switchport trunk native vlan 99 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100 switchport nonegotiate no shutdown ! end write memory </pre>